

Crystalline Lithium Imidazolate Covalent Organic Frameworks with High Li-Ion Conductivity

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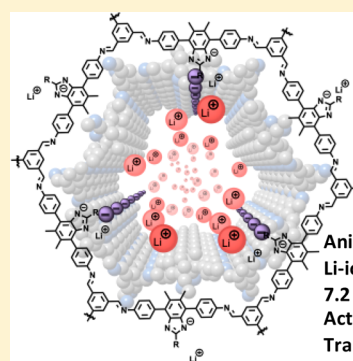
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Supporting Information

ABSTRACT: Ionic covalent organic frameworks (ICOFs) have recently emerged as promising candidates for solid-state electrolytes. Herein, we report the first example of a series of crystalline imidazolate-containing ICOFs as single-ion conducting COF solid electrolyte materials, where lithium cations freely travel through the intrinsic channels with outstanding ion conductivity (up to $7.2 \times 10^{-3} \text{ S cm}^{-1}$) and impressively low activation energy (as low as 0.10 eV). These properties are attributed to the weak Li ion–imidazolate binding interactions and well-defined porous 2D framework structures of such ICOFs. We also investigated the structure–property relationship by varying the electronic properties of substituents (electron donating/withdrawing) that covalently attached to the imidazolate groups. We found electron-withdrawing substituents significantly improve the ion-conducting ability of imidazolate-ICOF by weakening ion-pair interactions. Our study provides a convenient bottom-up approach toward a novel class of highly efficient single-ion conducting ICOFs which could be used in all solid-state electrolytic devices.



Anionic covalent frameworks
Li-ion Conductivity (rt):
 $7.2 \times 10^{-3} \text{ S cm}^{-1}$
Activation energy: 0.10 eV
Transference number: 0.81–0.93

INTRODUCTION

Imidazolate is the conjugate base of imidazole, which has been widely used as a chelating ligand in metal organic frameworks (MOFs).¹ One of these well-known examples is zeolitic imidazolate frameworks (ZIFs) where the transition metal ions, e.g., Fe, Co, Cu, and Zn, were connected by imidazolate linkers to form crystalline porous solids.^{1,2} Less commonly explored ZIF materials include lithium and lithium–boron nodes, which have been reported to form highly crystalline porous frameworks upon complexation with imidazolate and other charge-complementary ligands.^{3,4} Although imidazolates have been frequently used in MOFs and other coordination chemistry, to the best of our knowledge, imidazolate-containing covalent organic frameworks have not been reported. We envisioned that if imidazolates can be introduced into the backbones of covalent organic frameworks (COFs),^{5–9} a new type of ionic COFs (ICOFs)^{10–13} with aligned anion centers can be obtained. A loose bonding between the imidazolate anions and lithium cations has been reported in the lithium imidazolate salt;¹⁴ hence, we anticipate good lithium ion conductivity of such imidazolate-based COFs (ImCOFs) and their use as Li-ion battery electrolytes.

Most commercial lithium-ion batteries utilize electrolytes composed of free lithium salts dissolved in organic carbonate solvents. These volatile nonaqueous solutions have remained virtually unchanged since Sony's debut of the commercial lithium ion battery (LIB) in 1991. The flammability of these electrolytes raises significant safety concerns which have to be resolved before their use in high energy batteries to power future personal electronics and electric vehicles. Therefore, safe alternatives of electrolytes are highly desired to usher in a new era of energy-storage devices.¹⁵ In this context, ICOFs, an emerging class of crystalline and porous network materials with charged backbone structures, provide a promising alternative as solid-state electrolytic materials. In ICOFs, anionic functional groups can be immobilized into the framework backbones via a bottom-up synthetic approach, thus preventing the anion mobility and harnessing conductivity solely stemming from Li-ion mobility. These types of single-ion conducting solid electrolytes are advantageous to lithium salt-doped framework electrolytes,^{13,16–19} where both cations and anions are mobile and contribute to the conductivity, leading

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to low transference number and undesired side reactions.²⁰ Herein, we designed and synthesized a series of single-ion conducting imidazolate ICOFs (Li-ImCOFs), which exhibit excellent room-temperature lithium-ion conductivity up to $7.2 \times 10^{-3} \text{ S cm}^{-1}$, low activation energy of as low as 0.10 eV, and high transference number of 0.81. Additionally, we report the substituent effect on the conductivity of Li-ImCOFs by varying the substituents (H, CH₃, CF₃) on the imidazolate backbones.

EXPERIMENTAL SECTION

Synthesis of H-ImCOF. A mixture of imidazole substituted diamine **1-H** (25 mg, 0.075 mmol), trialdehyde **2** (8 mg, 0.05 mmol), ethanol/mesitylene (0.8 mL/0.2 mL) and an aqueous acetic acid solution (3 M, 0.2 mL) was added in an ampule. The tube was flash frozen at 77 K in liquid nitrogen and evacuated to an internal pressure of 200 mTorr. Then the tube was sealed and heated at 120 °C for 3 days without any disturbance. The pale-yellow precipitate was collected by vacuum filtration, washed with methylene chloride, ethanol, and acetone, and dried under reduced pressure. After further purification by Soxhlet extraction in THF for 24 h, **H-ImCOF** was obtained as a yellow solid (26 mg, 85%). Anal. Calcd for (C₂₇H₂₀N₄)_n: C, 81.0; H, 5.0; N, 14.0. Found: C, 77.1; H, 5.6; N, 12.9.

Synthesis of H-Li-ImCOF. To a well-dispersed suspension of **H-ImCOF** (20 mg) in hexane (1 mL) was added *n*-BuLi solution in hexane (1.6 M, 0.1 mL) dropwise in an ice bath over 10 min. The mixture was then stirred at room temperature for 16 h. The solid was isolated by centrifugation. Dry hexane (10 mL) was added, and the suspension was mixed well before centrifuging for 3–5 min. The hexane layer was decanted, and the solid was subjected to another round of washing. The washing cycle was repeated 5 times. After drying under vacuum, the product was collected as a yellow solid (19 mg, 94%). Anal. Calcd for (C₂₇H₁₉N₄Li)_n: C, 79.8; H, 4.7; Li, 1.7; N, 13.8. Found: C, 72.4; H, 5.4; Li, 3.2; N, 9.2.

CH₃-ImCOF (63%), **CH₃-Li-ImCOF** (90%), **CF₃-ImCOF** (72%), and **CF₃-Li-ImCOF** (91%) were synthesized following a procedure similar to that described above for **H-ImCOF** and **H-Li-ImCOF**. Anal. Calcd for **CH₃-ImCOF** (C₂₈H₂₂N₄)_n: C, 81.1; H, 5.4; N, 13.5. Found: C, 76.3; H, 5.3; N, 11.5; **CF₃-ImCOF** (C₂₈H₁₉N₄F₃)_n: C, 71.8; H, 4.1; N, 12.0. Found: C, 69.8; H, 4.1; N, 10.9; **CH₃-Li-ImCOF** (C₂₈H₂₁N₄Li)_n: C, 80.0; H, 5.0; Li, 1.7; N, 13.3. Found: C, 75.7; H, 4.8; Li, 2.4; N, 12.9; **CF₃-Li-ImCOF** (C₂₈H₁₈N₄F₃Li)_n: C, 70.9; H, 3.8; Li, 1.5; N, 11.8. Found: C, 67.1; H, 4.7; Li, 1.4; N, 10.1.

Typical Procedure for Ionic Conductivity Measurements. The samples were made by lightly compressing a Li-ImCOF pellet between blocking titanium electrodes in our custom all-solid-state cell. In a typical procedure, ~60 mg of the sample was loaded into a 13 mm die and pressed into a pellet (density ~0.21 g/cm³). Propylene carbonate (PC, 12 μL) was added to plasticize the pellet. The PC loading was estimated to be ~20 wt % according to the TGA analysis of the PC-solvated sample, which show ~20% weight loss below 200 °C. Electrochemical impedance spectroscopy (EIS) measurements were taken using a Solartron 1280C. Tests were performed over a frequency range of 1 MHz to 1 Hz with an oscillating voltage of 10 mV. The pellet's resistance was determined as the extrapolated high frequency intercept with the real x -axis of the Nyquist plot. Conductivity was calculated using eq 1, where L and A represent the pellet's thickness and cross-sectional area, respectively, while R is the measured resistance.

$$\sigma = \frac{L}{R \times A} \quad (1)$$

The material's activation energy was determined by taking EIS measurements over a range of temperatures. These tests were performed during a heating process within a small box furnace, allowing the temperature to equilibrate for 1 h between scans. The activation energy could be directly determined by measuring the slope of the Arrhenius plot of the resulting conductivity values.

$$t_{\text{Li}^+} = \frac{I_s(\Delta V - I_0 R_0)}{I_0(\Delta V - I_s R_s)} \quad (2)$$

The Bruce–Vincent–Evans (BVE) technique was used to calculate the lithium-ion transference number. A symmetric cell was constructed by applying Li-metal electrodes to both sides of a Li-ImCOF pellet. The BVE technique (eq 2) involves the application of a constant DC polarization (ΔV) so that the initial (I_0) and steady-state (I_s) current values can be measured. For this experiment, steady state was defined as a <1% change in current measured over a 10 min interval. Initial and steady-state resistance values, R_0 and R_s , were measured with EIS directly before and after the application of the DC polarization (ΔV).

Cyclic voltammetry (CV) and linear sweep voltammetry (LSV) were conducted on asymmetric (Ti/Li-ImCOF/Li) cells using a Solartron 1280C. To collect the cyclic voltammogram, a cyclic room temperature potential sweep (10 mV s⁻¹) was executed between -0.5 and 5 V in a dry argon environment. The linear sweep voltammograms were collected by first sweeping the voltage (1 mV s⁻¹) from the cell's open circuit potential (OCP) down to 0.2 V, followed by a secondary sweep from the OCP up to 6 V.

A symmetric Li/Li-ImCOF/Li cell was assembled to demonstrate the lithium metal compatibility and cyclability of the Li-ImCOF. Room temperature galvanostatic deposition/stripping of the Li-symmetric cell was performed using an Arbin B2000 battery testing station with a constant current density of 0.1 mA/cm².

RESULTS

Synthesis and Characterization of COFs. In a typical procedure, a mixture of the imidazole-containing diamine **1** (**1-H**, **1-CH₃**, **1-CF₃**) and 1,3,5-triformylbenzene **2** in ethanol, mesitylene, and aqueous acetic acid was heated at 120 °C for 3 days without stirring. A pale yellow powder precipitated, which was collected and purified by washing with methylene chloride, ethanol, and acetone followed by Soxhlet extraction (THF) of the other remaining impurities to provide the product ImCOFs (63–85%). Three ImCOFs, **H-ImCOF**, **CH₃-ImCOF**, and **CF₃-ImCOF**, with different substituents ($R = \text{H, CH}_3, \text{ and CF}_3$) on the imidazole ring were synthesized. Lithium imidazolate frameworks, **H-Li-ImCOF**, **CH₃-Li-ImCOF**, and **CF₃-Li-ImCOF**, were obtained by treating the ImCOFs with *n*-BuLi in hexane solution for 16 h (Figure 1).

ImCOFs and Li-ImCOFs were characterized by solid-state NMR spectroscopy, IR spectroscopy, powder X-ray diffraction (PXRD), transmission electron microscopy (TEM), TGA, and N₂ adsorption. In ¹³C cross-polarization magic-angle spinning (CP-MAS) NMR spectra of ImCOFs, we observed ¹³C peaks around 159–160 ppm corresponding to the carbon signals of imine carbon. In addition, aldehyde carbon peaks around 190–200 ppm are absent in the ¹³C NMR spectra of all three ImCOFs (Figures S1, S3, and S5). The ¹³C solid-state NMR spectra of Li-ImCOFs are similar to those of parent ImCOFs (Figures S2, S4, S6). ⁷Li solid-state NMR spectra of three Li-ImCOFs show a singlet at 1.65, 1.33, and -0.29 ppm, respectively, for **CH₃-ImCOF**, **H-ImCOF**, and **CF₃-ImCOF** (Figures S7–S9). The Li-ion loadings in **H-Li-ImCOF**, **CH₃-Li-ImCOF**, and **CF₃-Li-ImCOF** were estimated to be 3.2%, 1.7%, 1.4%, respectively, according to ICP-MS analysis. The Fourier transform infrared (FT-IR) spectra of ImCOFs show strong imine (C=N) peaks at 1624–1625 cm⁻¹ (Figure S10–S12).

The crystallinities of ImCOFs and Li-ImCOFs were evaluated by powder X-ray diffraction (PXRD) measurements (Figure 2 and Figures S13 and S14). All of the ImCOFs and Li-ImCOFs have good crystallinity, showing an intense peak at

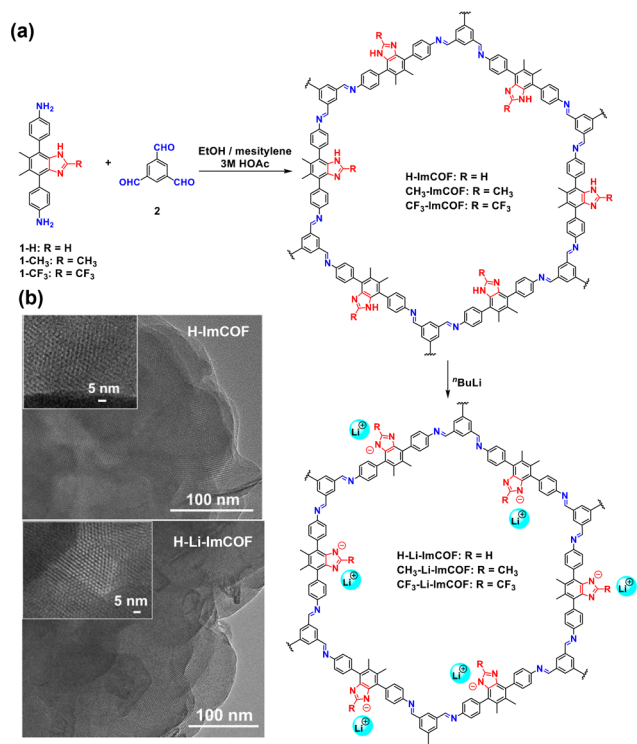


Figure 1. (a) Synthesis of ImCOFs and Li-ImCOFs; (b) TEM images of H-ImCOF and H-Li-ImCOF.

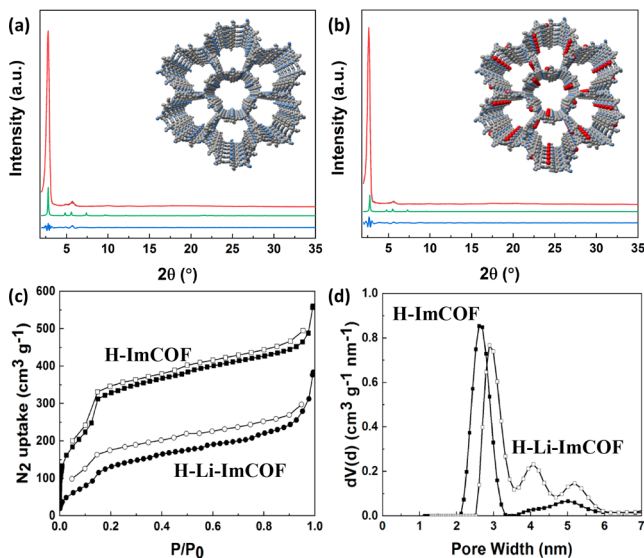


Figure 2. (a, b) PXRD patterns of H-ImCOF (a) and H-Li-ImCOF (b): observed (red), refined modeling profile (black), simulated (green), and difference between observed PXRD and refined modeling pattern (blue). Inset: the top views of AA-stacking model of H-ImCOF and H-Li-ImCOF (gray, C; blue, N; red, Li). (c) Nitrogen adsorption–desorption isotherms of H-ImCOF (square) and H-Li-ImCOF (circles). Adsorption and desorption data points are represented by filled and empty symbols, respectively. (d) Pore size distribution of H-ImCOF (filled square) and H-Li-ImCOF (empty square).

$2\theta = 2.5\text{--}3.5^\circ$ and a small peak at $2\theta = 4.9\text{--}5.5^\circ$, which correspond to 100 and 110 plane reflections, respectively. For example, the experimental PXRD patterns (red) of H-ImCOF show a set of intensive peaks at $2\theta = 2.8^\circ$, 4.9° , and 5.7°

corresponding to planes 100, 110, and 200 respectively. Unit cell parameters of H-ImCOF were refined by the Pawley method to be $\alpha = \beta = 90^\circ$, $\gamma = 120^\circ$, $a = b = 36.95$, $c = 4.15$ Å (Table S1). Crystal lattice packing of H-ImCOF was determined to be fully eclipsed AA stacking mode with an interlayer distance of 4.15 Å. The simulated PXRD patterns match well with the experimental results. After lithiation, the H-Li-ImCOF maintained the crystalline structure in all three dimensions, showing the diffraction peaks at $2\theta = 2.6^\circ$ and 5.6° corresponding to 100 and 110 planes, respectively. (Figure 2b). Similarly, PXRD measurements confirmed the AA stacking modes of highly crystalline frameworks of CH₃-ImCOF and CF₃-ImCOF and their lithiated analogues Li-CH₃-ImCOF and Li-CF₃-ImCOF (Figures S13 and S14). Transmission electron microscopy (TEM) images of H-ImCOF and H-Li-ImCOF reveal the nanoscale morphology of individual COF crystallites (Figure 1b). Domains oriented with their *ab* plane perpendicular to the viewing direction show the highly ordered hexagonal arrangement of the mesopores around 2.6 nm, which matches well with pore size distribution measurements described below.

Permanent porosities of ImCOFs and Li-ImCOFs were measured by nitrogen adsorption isotherm measurements at 77 K. The type IV isotherm was observed for H-ImCOF with the Brunauer–Emmett–Teller (BET) surface area of $1120\text{ m}^2\text{ g}^{-1}$ and pore volume of $0.87\text{ cm}^3\text{ g}^{-1}$. After lithiation, H-Li-ImCOF shows a much decreased BET surface area of $350\text{ m}^2\text{ g}^{-1}$ and a pore volume of $0.59\text{ cm}^3\text{ g}^{-1}$. The pore size distribution calculated by using quenched solid density functional theory (QSDFT) and cylindrical pore model shows a pore size distribution mainly at 2.6 and 2.9 nm, respectively, for H-ImCOF and H-Li-ImCOF, which match well with their theoretical pore size ~ 3.0 nm. Methyl- or trifluoromethyl-substituted derivatives, CH₃-ImCOF and CF₃-ImCOF, show considerably lower BET surface areas of 460 and $705\text{ m}^2\text{ g}^{-1}$, respectively (Figures S16 and S17). The pore size distribution analyses of CH₃-ImCOF and CF₃-ImCOF show smaller major pores (2.3 and 2.2 nm, respectively) compared to that of H-ImCOF (Figures S16 and S17).

The lithiated COFs show good stability during in-air handling. The ImCOFs were subjected to a thorough cleaning process, which involves washing the framework products with 10 times volume of fresh dry hexane for five times to remove extra *n*-BuLi. The Li elemental analysis showed there are no large excess Li ions left in the lithiated COFs. We did not observe any change in their color, morphology, and spectra after extensive characterization including PXRD, ssNMR, or IR under ambient conditions. Thermogravimetric analyses (TGA) show that all three ImCOFs have excellent thermal stability with onset decomposition temperatures around 450°C . The lithiated samples, H-Li-ImCOF, CH₃-Li-ImCOF, and CF₃-Li-ImCOF, exhibit slightly lower thermal stabilities showing continuous weight loss of 8%, 15%, and 15%, respectively, from room temperature to 450°C (Figure S18).

Next, we investigated the lithium ion conductivities of Li-ImCOFs to evaluate their potential applications as solid electrolyte materials.²¹ The electrochemical analysis was performed on dense Li-ImCOFs pellets pressed within custom polyether ether ketone (PEEK) lined 13 mm diameter all-solid-state cell dies. Prior to analysis, Li-ImCOFs were vacuum-dried overnight at 120°C to remove any residual moisture. The samples were then solvated with propylene carbonate (PC, $\sim 20\text{ wt } \%$), and their ionic conductivities were

measured with electrochemical impedance spectroscopy (EIS). After PC uptake, the lithium ion loadings were 2.7%, 1.4%, and 1.1% in **H-Li-ImCOF**, **CH₃-Li-ImCOF**, and **CF₃-Li-ImCOF**, respectively. Nyquist plots of **H-Li-ImCOF** (Figure 3a) display

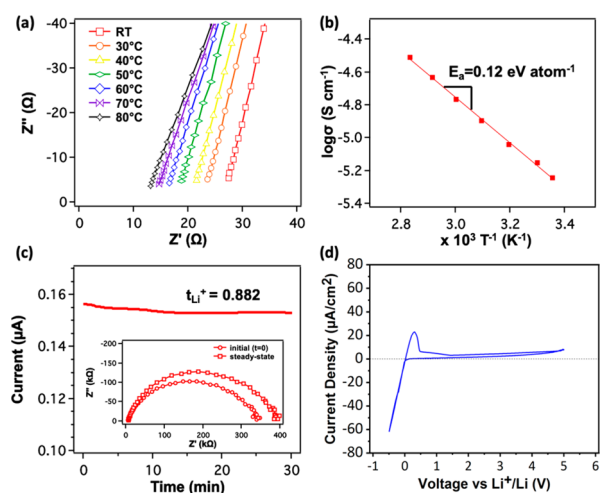


Figure 3. Electrochemical measurements of **H-Li-ImCOF** pellets solvated with PC: (a) Nyquist plots of electrochemical impedance spectroscopy (EIS) measurements made over a range of temperatures; (b) Arrhenius plot of ionic conductivity as a function of temperature; (c) lithium transference number calculated using the Bruce–Vincent–Evans technique; (d) cyclic voltammogram of the Ti/**H-Li-ImCOF**/Li cell.

steep spikes with no trace of the high frequency semicircles characteristic of EIS. Because no semicircular impedance responses were observed, individual resistance contributions (grain boundary and bulk) could not be differentiated. Instead, the net resistance, determined by extrapolating the electrode spike to the “real” x -axis of the Nyquist plot, is used to calculate the material’s overall conductivity. With this approach, a room temperature conductivity of $5.3 \times 10^{-3} \text{ S cm}^{-1}$ was calculated for the solvated **H-Li-ImCOF** pellet. Similar sample preparation and measurement procedures were applied to evaluate the ion conductivities of **CH₃-Li-ImCOF** and **CF₃-Li-ImCOF**. According to the Nyquist plots at room temperature, the ion conductivities of 7.2×10^{-3} and $8.0 \times 10^{-3} \text{ S cm}^{-1}$ were determined for **CF₃-Li-ImCOF** and **CH₃-Li-ImCOF**, respectively. Li-ion-conducting activation energies were evaluated by analyzing a set of Nyquist plots measured at various temperatures ranging from room temperature to 85 °C. As shown in Figure 3b, the linear Arrhenius plot was obtained for **H-Li-ImCOF**, from which the activation energy was calculated to be 0.12 eV. The activation energies of **CF₃-Li-ImCOF** and **CH₃-Li-ImCOF** samples were obtained through similar methods and determined to be 0.10 and 0.27 eV, respectively. Average Li⁺ transference numbers were determined by the BVE method. High Li⁺ transference numbers, 0.88, 0.93, and 0.81, were calculated for **H-Li-ImCOF**, **CH₃-Li-ImCOF**, and **CF₃-Li-ImCOF**, respectively (Figure 3c and Figure S20). Linear sweep voltammetry (LSV) was used to measure the electrochemical stability windows of Li-ImCOFs. The linearly sloping profile of the anodic and cathodic sweeps between 0.2 to 4.5 V and steep increase in measured current above 4.5 V were observed for **H-Li-ImCOF** (Figures S21). LSV graphs of **CH₃-Li-ImCOF** and **CF₃-Li-ImCOF** showed some reaction occurring around 1.0–2.0 V. The cyclic

voltammogram (Figure 3d) of the **H-Li-ImCOF** displays a wide window of electrochemical stability, as no significant current flow corresponding to electrolyte decomposition was observed up to 5 V. Only cathodic and anodic currents corresponding to lithium deposition ($\text{Li}^+ + \text{e}^- \rightarrow \text{Li}$) and stripping ($\text{Li} \rightarrow \text{Li}^+ + \text{e}^-$) were observed near 0 V vs Li. Linear sweep voltammograms of the remaining Li-ImCOFs (Figures S21) show that they are fairly stable up to ~4.5 V. Figures S22 and S23 display the voltage profiles of the galvanostatically cycled Li/**H-Li-ImCOF**/Li- and Li/**CF₃-Li-ImCOF**/Li-symmetric cells. These cells achieved stable cyclability at room temperature, confirming the potential of Li-ImCOFs as solid electrolyte materials.

DISCUSSION

The stability of imidazolate anions is affected by the electronic properties of the substituents. Imidazoles substituted with more electron-withdrawing groups are more acidic; thus, their corresponding imidazolate anions form loose ion pairs with lithium cations, leading to higher lithium ion conductivity. We chose CH₃, H, and CF₃ substituents on imidazole moieties to study the substituent effects. ⁷Li solid-state NMR spectra of **H-Li-ImCOF**, **CH₃-Li-ImCOF**, and **CF₃-Li-ImCOF** clearly showed a single ⁷Li resonance peak, indicating the uniform chemical environment of Li ions in each of these ICOFs (Figures S7–S9). However, the ⁷Li resonance signals of **H-Li-ImCOF**, **CH₃-Li-ImCOF**, and **CF₃-Li-ImCOF** are shifted from 1.65 to 1.33 and −0.29 ppm when the substituents are varied from electron-donating CH₃ to H and electron-withdrawing CF₃. This high-field shift of ⁷Li NMR signals is due to the lengthening of the Li–N distance and the diminished solid-state ion pairing, which is most pronounced in the case of **CF₃-Li-ImCOF**.^{16,22} This is consistent with our molecular modeling calculations on three simple model compounds, showing the increase of Li–N bond length with the increasing electronegativity ($\text{CH}_3 < \text{H} < \text{CF}_3$) of the substituents on the imidazolate ring (Figure S24). The half line width of ⁷Li NMR signal of **CF₃-Li-ImCOF** (347 Hz) was also considerably narrower compared to those of **H-Li-ImCOF** (512 Hz) and **CH₃-Li-ImCOF** (546 Hz). These results suggest loose ion pairing and higher lithium ion mobility in **CF₃-Li-ImCOF**, which agrees well its higher lithium ion conductivity.

Although all three Li-ImCOFs are crystalline, the degree of crystallinities of **H-Li-ImCOF** and **CF₃-Li-ImCOF** are notably higher than that of **CH₃-Li-ImCOF** (Figure 2b and Figures S13 and S14). PXRD spectra of **CH₃-Li-ImCOF** show a broad hump in the 2θ range of 15°–30°, which suggests the presence of amorphous phase. In addition, the diffraction peak corresponding to the 100 plane of **CH₃-Li-ImCOF** is not as sharp as that of **H-Li-ImCOF** or **CF₃-Li-ImCOF**. The BET surface area of **CH₃-Li-ImCOF** was also the lowest of the three lithiated COFs. Although the PXRD patterns of Li-ImCOFs after lithiation remain mostly the same as those of parent ImCOFs, their surface areas are significantly decreased presumably due to the inevitable damages on the pore structure during the postfunctionalization process.²³ Similarly, the changes of pore volume and size observed after lithiation in **H-Li-ImCOF** and **CH₃-Li-ImCOF** are likely due to a bit disruption of the original framework structures during the postfunctionalization. Little change of pore volume and size was observed for **CF₃-Li-ImCOF** after lithiation, suggesting its higher structural stability. Although the surface area might not

Table 1. Summary of the Typical Performance of Inorganic Sulfide Electrolyte, MOF-, and COF-Based Li⁺ Solid-State Electrolyte

material	description			BET m ² /g	conductivity (S· cm ⁻¹) ^a	activation energy (eV)	transference no.	ref
	backbone	Li salt	plasticizer					
Li ₆ PS ₄ Cl					4.6 × 10 ⁻⁴	0.33	1.00	25
Li ₁₀ GeP ₂ S ₁₂					1.2 × 10 ⁻²	0.25	1.00	26
Li ₇ P ₃ S ₁₁					4.1 × 10 ⁻³	0.15	1.00	27
Al-Td-MOF-1	anionic tetrahedral AlO ₄ site		~50 wt % PC	656	5.7 × 10 ⁻⁵	0.11		28
MIT-20-LiCl	anionic MOF backbone		~70 wt % PC		1.3 × 10 ⁻⁵	0.32	0.66	29
LiClO ₄ @COF-5	COF-5	LiClO ₄			2.6 × 10 ⁻⁴	0.37		19
ICOF-2	anionic spiroborate ICOF		~55 wt % PC	1259	3.1 × 10 ⁻⁵	0.24	0.80	10
CD-COF-Li/LiPF ₆ -EC- DMC	cyclodextrin-based ICOF	LiPF ₆	EC + DMC	760	2.7 × 10 ⁻³	0.26		13
Li-CON-TFSI	TFSI ⁻ /cationic CON	LiTFSI		118	5.7 × 10 ⁻⁵	0.34	0.61	16
Li+@TPB-BMTP-COF	2D PEO-based COF	LiClO ₄			6.0 × 10 ⁻⁶ (40 °C)	0.87		17
lithium imidazolate 4	small molecule				7.2 × 10 ⁻⁶			
H-ImCOF and LiClO ₄ ^b	neutral H-ImCOF	LiClO ₄	~20 wt % PC		4.0 × 10 ⁻⁵		0.21	Figure S20
H-Li-ImCOF	Li ⁺ /anionic ICOF		~20 wt % PC	350	5.3 × 10 ⁻³	0.12	0.88	This work
CH₃-Li-ImCOF	Li ⁺ /anionic ICOF		~20 wt % PC	328	8.0 × 10 ⁻⁵	0.27	0.93	This work
CF₃-Li-ImCOF	Li ⁺ /anionic ICOF		~20 wt % PC	633	7.2 × 10 ⁻³	0.10	0.81	This work

^aConductivity at room temperature. ^bSame Li-ion loading as **H-Li-ImCOF**.

influence the lithium ion conductivity, the crystallinity and the presence of well-ordered channels in electrolyte materials would facilitate the migration of lithium ions thus enhance their conductivity.²⁴ This notion is consistent with our experimental results, which show much higher lithium ion conductivity in more ordered **H-Li-ImCOF** and **CF₃-Li-ImCOF** electrolytes compared to that in **CH₃-Li-ImCOF** with lower crystallinity.

We found as-prepared Li-ImCOF pellets are modest Li-ion conductors (~5 × 10⁻⁷ S cm⁻¹). However, upon solvation with PC, the conductivity of Li-ImCOFs increases by several orders of magnitude. Propylene carbonate is a polar solvent of high dielectric constant and does not directly contribute to the conductivity. Rather, it facilitates the weakly bound Li ions to dissociate from the insoluble Li-ImCOF skeleton, allowing them to quickly travel through the large channels present in the long-range ordered crystal structure.^{24,29,30} Maximum conductivities were achieved upon solvating the Li-ImCOFs with only one drop of PC, constituting ~20 wt % of the COF pellet's mass. By solvating with such a small amount of PC, we were able to maintain a robust, freestanding Li-ImCOFs pellet and measure the unimpeded flow of Li ions over a wide range of temperatures. This is in contrast to many other reported works that rely on solvent loadings exceeding 90 wt % to achieve appreciable conductivity from similar organic electrolytes.^{31–33} The amount of PC added to Li-ImCOFs is almost insignificantly small, which should pose little to no flammability threat, especially in comparison to the conventional liquid electrolyte cells (such as dissolving the Li salt in PC solution) currently on the market. Besides, the activation energy measurements (Figure 3a,b) show no negative impact on ionic conductivity upon pro-longed heating up to 80 °C, indicating its safety at high working temperature. Furthermore, the auto ignition temperature of PC is ~455 °C, meaning that combustion due to elevated cycling temperature would not

occur. It is possible that PC vapor could ignite at lower temperatures, but in a sealed cell, this is extremely unlikely and would require external flame source and would again be expected to be minimal or inconsequential due to the small amount of PC included in the mostly solid electrolyte.

The utilization of solvating additives can have a large impact on the mechanical properties of organic electrolytes and lead to substantial electrochemical, chemical, and thermal instabilities. This is not the case for our free-standing, solvated Li-ImCOF electrolytes, as the CV plot in Figure 3d clearly shows no current flow associated with electrolyte instability or breakdown up to 5 V, while the activation energy measurements (Figure 3a,b) show no negative impact on ionic conductivity upon prolonged heating up to 80 °C, indicating its safety at high working temperature. Our case is also advantageous to many previous reports on conductive organic porous polymer which have relied upon the addition of large mass fractions (>30 wt %) of polymeric binders to prepare freestanding films.³⁴ While these composites are mechanically robust, their uneven distribution of conductive polymer powders negatively impedes the movement of Li ions through the films. To the best of our knowledge, the ionic conductivities of **H-Li-ImCOF** (5.3 × 10⁻³ S cm⁻¹) and **CF₃-Li-ImCOF** (7.2 × 10⁻³ S cm⁻¹) are the highest ever recorded for a freestanding polymer-based solid electrolyte (Table 1). They are more than two orders magnitude higher than the lithium ion conductivity of spiroborate-based ICOF (3 × 10⁻⁵ S cm⁻¹) measured under similar conditions in the absence of additional lithium salts.¹⁰ Even in the presence of external lithium salts, such as LiClO₄, LiPF₆, or LiTFSI, the conductivities of many COFs, e.g., Li-CON-TFSI,¹⁶ Li+@TPB-BMTP-COF,¹⁷ CD-COF-Li/LiPF₆-EC-DMC,¹³ or LiClO₄@COF-5,¹⁹ are 2–1200 times lower than those of **H-Li-ImCOF** and **CF₃-Li-ImCOF**. The ambient conductivities of the solvated **H-Li-ImCOF** and **CF₃-Li-ImCOF** even surpass

those of the heavily researched inorganic sulfide solid electrolyte crystal structures; e.g., monoclinic binary $\text{Li}_2\text{S}-\text{P}_2\text{S}_5$ thio-LISICON ($1.3 \times 10^{-3} \text{ S cm}^{-1}$),³⁵ $\text{Li}_6\text{PS}_5\text{X}$ Argryrodites ($1 \times 10^{-3} \text{ S cm}^{-1}$),³⁶ superionic $\text{Li}_7\text{P}_3\text{S}_{11}$ ($4.1 \times 10^{-3} \text{ S cm}^{-1}$),³⁷ and the $\text{Li}_{10}\text{MP}_2\text{S}_{12}$ family (LiGePS $2.1 \times 10^{-3} \text{ S cm}^{-1}$, LiSiPS $2.3 \times 10^{-3} \text{ S cm}^{-1}$, and LiSnPS $4.0 \times 10^{-3} \text{ S cm}^{-1}$).³⁸ In fact, only liquid electrolytes based on lithium salts (e.g., lithium perchlorate, LiClO_4) dissolved in volatile organic carbonates, e.g., ethylene carbonate (EC), propylene carbonate (PC), diethyl carbonate (DEC), or ionic liquids (e.g., $\text{PYR}_{13}\text{FSI}$), have been shown to have higher conductivities than $\sim 1 \times 10^{-2} \text{ S cm}^{-1}$ at room temperature.^{39,40}

This electrochemical stability and large effective conductivity allowed a $\text{Li}/\text{H-Li-ImCOF}/\text{Li}$ symmetric cell to achieve >40 galvanostatic deposition/stripping cycles at room temperature (Figure S22). The stability of the cell's voltage profile confirms the Li-ImCOF's compatibility with metallic lithium and its ionic conductivity estimated from direct current (DC) polarization are on the same order of magnitude as that obtained from room-temperature EIS measurements ($\sim 1 \times 10^{-3} \text{ S cm}^{-1}$).

The methyl-substituted analogue, $\text{CH}_3\text{-Li-ImCOF}$, shows lithium conductivity of $8.0 \times 10^{-3} \text{ S cm}^{-1}$, which is a 70–90-fold decrease compared to those of H-Li-ImCOF and $\text{CF}_3\text{-Li-ImCOF}$. The observed order of conductivity, $\text{CF}_3\text{-Li-ImCOF} > \text{H-Li-ImCOF} > \text{CH}_3\text{-Li-ImCOF}$, matches well with the strength of imidazolate and lithium ion pairing. In $\text{CF}_3\text{-Li-ImCOF}$, the negative charge on the nitrogen is largely delocalized due to the strong electron-withdrawing effect of CF_3 substituent, leading to weaker ion pairing and higher ion conductivity. On the contrary, the CH_3 substituent is electron donating and, thus, less effective at distributing the negative charge, resulting in greater ion pair strength and low lithium ion conductivity. The low crystallinity and porosity of $\text{CH}_3\text{-Li-ImCOF}$ might also contribute to its inferior ion conducting property to some extent. Our study clearly shows that the ion conductivities of imidazolate-based ICOFs can be modulated by tuning electronic properties of the imidazole substituents, which directly affect imidazolate–lithium ion-pair formation and lithium-ion mobility.

Although high frequency semicircles were observed in the Nyquist plots of solvated $\text{CH}_3\text{-Li-ImCOF}$, they were absent for solvated H-Li-ImCOF and $\text{CF}_3\text{-Li-ImCOF}$ under a similar testing temperature and frequency range. Besides their low resistance, the steep, nearly vertical “electrode spikes” observed for H-Li-ImCOF and $\text{CF}_3\text{-Li-ImCOF}$ are also indicative of the capacitive behavior of an ionically conducting electrolyte intimately interfacing with a blocking metal (titanium) electrode. In fact, the close similarity of the H-Li-ImCOF and $\text{CF}_3\text{-Li-ImCOF}$ pellets' AC response to the perfectly vertical ideal capacitive spike indicates that a smooth, dense, conformal contact has been made with the blocking electrodes after a simple cold pressing.⁴¹

The linearity of Arrhenius plots observed for all three Li-ImCOFs indicate that Li-ion conduction in these ICOFs is decoupled from its negatively charged framework. This is uncommon, as the relationship between conductivity and temperature for solid organic and liquid electrolytes typically deviates from linearity at elevated temperatures, a phenomenon that can be closely approximated by the well-known Vogel–Tammann–Fulcher (VTF) equation.^{42,43} This difference from the conductivity response of an organic liquid electrolyte indicates that the Li ions loosely bound to the Li-ImCOFs

framework are not simply dissolving and diffusing through the solvating PC. Instead, the Arrhenius nature of the Li-ImCOFs suggests that conduction through the porous channels of its rigid framework occurs through a mechanism similar to the hopping of Li ions between lattice sites in an inorganic superionic crystal. Unlike closely packed inorganic crystal structures, where Li ions have to squeeze between high energy environment bottlenecks to percolate through torturous networks or unoccupied lattice sites, the Li ions in the Li-ImCOFs have long, straight, wide open tunnels to speed through. These large conduction channels in Li-ImCOFs in combination with the lubricating effect of the solvating PC result in their low activation energy of 0.10–0.27 eV. The order of activation energy of Li-ImCOFs agrees well with the trend of their lithium ion conductivity order: $\text{CF}_3\text{-Li-ImCOF}$ with the highest lithium ion conductivity shows the lowest activation energy of 0.10 eV; $\text{CH}_3\text{-Li-ImCOF}$ with lowest conductivity shows the highest activation energy of 0.27 eV; the activation energy of H-Li-ImCOF is 0.12 eV (Figure 4).

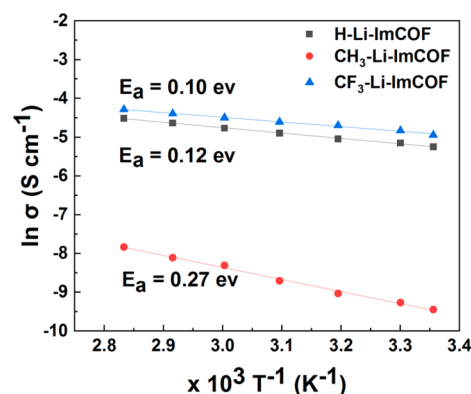


Figure 4. Arrhenius plot of ionic conductivity of H-Li-ImCOF , $\text{CH}_3\text{-Li-ImCOF}$, and $\text{CF}_3\text{-Li-ImCOF}$ as a function of temperature.

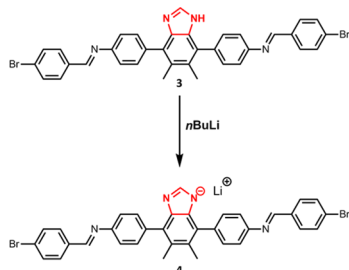
The activation energy of $\text{CF}_3\text{-Li-ImCOF}$ or H-Li-ImCOF is half of the activation energy recently reported for a similar spiroborate-based COF (0.24 eV)¹⁰ and even smaller than the best inorganic superionic crystals ($\sim 0.15 \text{ eV}$).^{25–27} This low activation energy is attributed to the weak ion pair strength in $\text{CF}_3\text{-Li-ImCOF}$ and H-Li-ImCOF and the well-delocalized negative charge of imidazolate anions as well as the enhanced lithium ion mobility through the long channels intrinsic to their highly ordered crystal structures.

All three Li-ImCOFs have negatively charged groups immobilized in their rigid backbone structures, thus only allowing the movement of the loosely bound lithium cations. As a result, these materials behave as novel single-ion conductors with transference numbers approaching unity (0.81–0.93). The small deviation of the transference numbers from unity is likely due to the unavoidable measurement errors and initial passivation reactions occurring between the Li metal and COF interface, which are common for battery materials in contact with Li and can most likely be attributed to PC “attacking” the lithium metal surface. These values are drastically higher than the transference numbers (0.2–0.5) reported for liquid and solid polymer electrolytes, such as PEO-based⁴² or COF-based electrolytes,^{13,16,17} that rely on the dissociation of unbound Li-salts for their charge carriers. These small transference numbers show that the large ionic conductivities reported for these electrolytes are primarily

attributed to the diffusion of their solvated anions. This means that the effective Li-ion conductivities of these electrolytes are in reality far lower than they appear. The presence of mobile cations and anions will also lead to the development of concentration gradients within these electrolytes, as the mobility of the positive and negative species is not equivalent, and the movement of anions will be blocked at the electrode interfaces. This separation of charge will lead to the polarization of the electrolyte resulting in the reduced performance of the cell. To further support the advantage of immobilizing the anions within the ICOF structure, we evaluated the performance of a physically blended ImCOF/Li-salt electrolyte system. To provide a fair comparison, we combined the unlithiated parent **H-ImCOF** with a conventional LiClO₄ salt such that the mixture's Li-ion loading would be equivalent to that of **H-Li-ImCOF**. Upon solvation with ~20 wt % PC, this mixture displayed a low lithium ion transference number (0.21) and an ionic conductivity (4×10^{-5} S/cm), which is almost 2 orders of magnitude lower than that of **H-Li-ImCOF** (Figure S20). These results further confirm the benefit of anion immobilization within the Li-ImCOF structure.

The excellent lithium ion conductivity is likely attributed to the loose-ion pairing between lithium and imidazolate anions immobilized in the backbone of polymer frameworks as well as their well-defined pore channels where lithium ions can freely transport. To support the importance of the interconnected framework structures of Li-ImCOFs, we prepared a small molecule imidazolate salt **4** (Scheme 1), whose structure

Scheme 1. Synthesis of Model Compound



closely resembles the Li-ImCOFs, and measured its lithium ion conductivity. We found the ion conductivity of the model compound under the same conditions is only 7.2×10^{-6} S cm⁻¹, which is around 1000 times less conductive than **H-Li-ImCOF** or **CF₃-Li-ImCOF**. This result indicates the framework structure is critical and provides intrinsic transport pathways which facilitate the Li-ion conduction in the reported solid-state electrolyte.

CONCLUSION

For the first time, we report the design and synthesis of imidazolate-based COFs and their properties as excellent single Li-ion-conducting solid electrolyte materials. Imidazolate-ICOFs were obtained as crystalline powders after a simple lithiation postfunctionalization of imidazole-COFs. While the parent COFs exhibit negligible conductivity, lithiated COFs show outstanding ionic conductivity values (up to 7.2×10^{-3} S cm⁻¹), low activation energy (as low as 0.10 eV), high transference number (up to 0.91), and wide electrochemical stability window (up to 4.5 V), good compatibility with Li-metal electrodes, and outstanding lithium deposition/stripping

cycling at room temperature. The structure–property relationship study conducted by varying the substituents on the imidazolate moieties shows significant substituent effects on the ion conductivity of Li-ImCOFs. Electron-withdrawing substituents (e.g., CF₃) help the delocalization of the negative charge of imidazolate anions, thus weakening the interactions between Li⁺ and the framework skeleton and enhancing Li⁺ mobility, leading to high ion conductivity. Clearly, imidazolate-COFs represent a new type of highly promising single-ion-conducting solid-state electrolytes which can be used in high-efficiency and stable Li-ion batteries.

ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge on the ACS Publications website at DOI: 10.1021/jacs.9b02448.

Detailed experimental procedures, FT-IR spectra, solid-state ¹³C CP-MAS NMR spectrum, solid-state ⁷Li NMR spectrum, gas adsorption data, TGA graph, SEM and TEM images, structural simulation, and Li-ion conductivity measurements (PDF)

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Notes

The authors declare no competing financial interest.

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